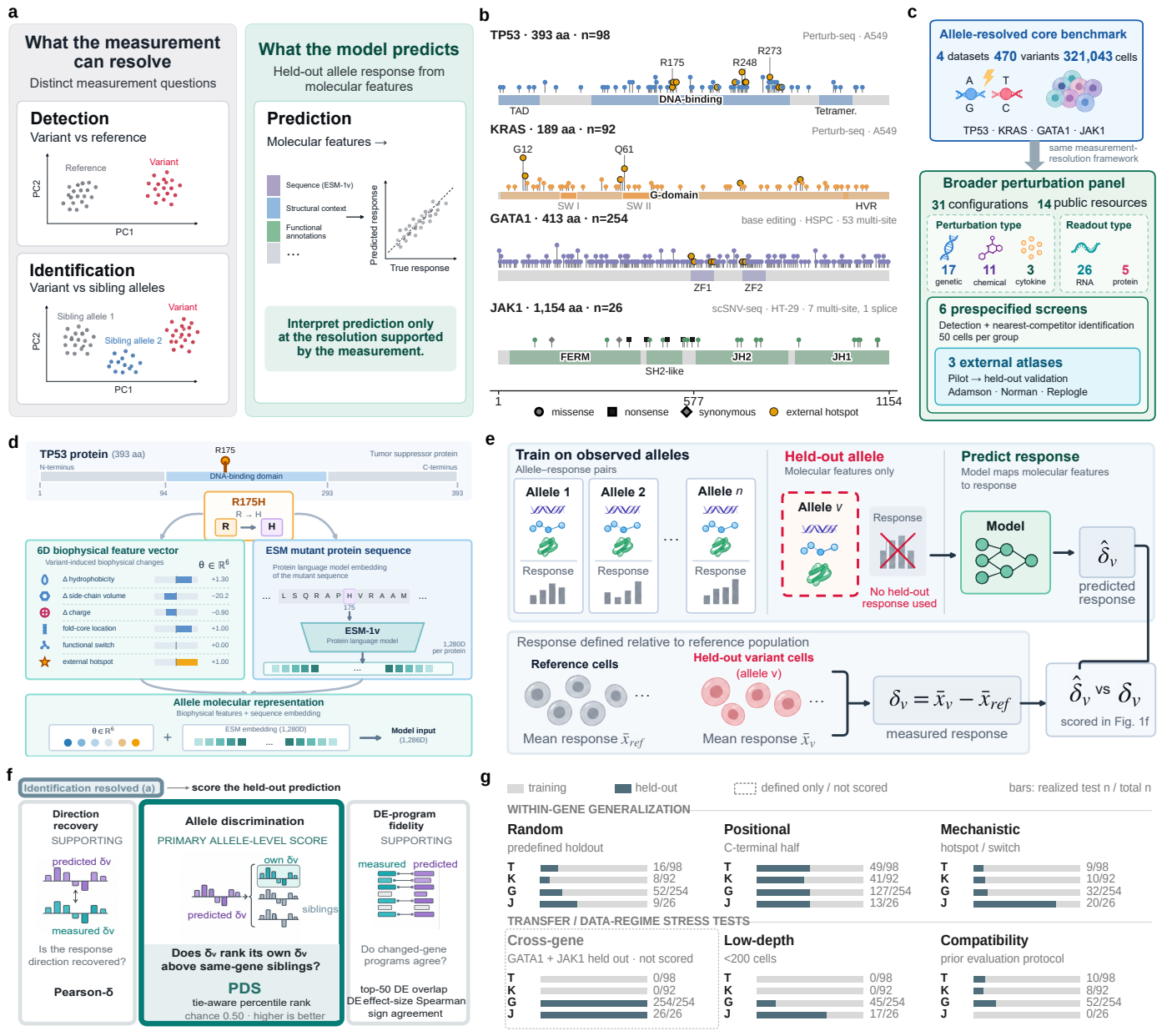

a
What the measurement can resolve
Distinct measurement questions
Detection
Variant vs reference
Reference
Variant
PC2
PC1
Identification
Variant vs sibling alleles
Sibling allele 1
Sibling allele 2
Variant
What the model predicts
Held-out allele response from molecular features
Prediction
Molecular features →
Sequence (ESM-1v)
Structural context
Functional annotations
Predicted response
True response
Interpret prediction only at the resolution supported by the measurement.
b
TP53 · 393 aa · n=98
Perturb-seq · A549
R175
R248
R273
DNA-binding
TAD
Tetramer.
KRAS · 189 aa · n=92
Perturb-seq · A549
G12
Q61
G-domain
SW I
SW II
HVR
GATA1 · 413 aa · n=254
base editing · HSPC · 53 multi-site
ZF1
ZF2
JAK1 · 1,154 aa · n=26
scSNV-seq · HT-29 · 7 multi-site, 1 splice
FERM
JH2
JH1
SH2-like
1
577
1154
missense
nonsense
synonymous
external hotspot
c
Allele-resolved core benchmark
4 datasets 470 variants 321,043 cells
A
T
G
C
TP53 · KRAS · GATA1 · JAK1
same measurement-resolution framework
Broader perturbation panel
31 configurations 14 public resources
Perturbation type
17 genetic
11 chemical
3 cytokine
Readout type
26 RNA
5 protein
6 prespecified screens
Detection + nearest-competitor identification
50 cells per group
3 external atlases
Pilot → held-out validation
Adamson · Norman · Replogle
d
TP53 protein (393 aa)
R175
Tumor suppressor protein
N-terminus
DNA-binding domain
C-terminus
1
94
293
393
R175H
R → H
R
H
6D biophysical feature vector
Variant-induced biophysical changes
θ ∈ ℝ⁶
Δ hydrophobicity
+1.30
Δ side-chain volume
−20.2
Δ charge
−0.90
fold-core location
+1.00
functional switch
+0.00
external hotspot
+1.00
ESM mutant protein sequence
Protein language model embedding of the mutant sequence
L S Q R A P H V R A A M
175
ESM-1v
Protein language model
1,280D per protein
Allele molecular representation
Biophysical features + sequence embedding
θ ∈ ℝ⁶
ESM embedding (1,280D)
Model input (1,286D)
e
Train on observed alleles
Allele–response pairs
Allele 1
Allele 2
Allele n
Response
Held-out allele
Molecular features only
Allele v
Response
No held-out response used
Predict response
Model maps molecular features to response
Model
predicted response
Response defined relative to reference population
Reference cells
Mean response $\bar{x}_{ref}$
Held-out variant cells (allele v)
Mean response $\bar{x}_v$
$\delta_v = \bar{x}_v - \bar{x}_{ref}$
measured response
$\hat{\delta}_v$ vs δ_v
scored in Fig. 1f
f
Identification resolved (a)
score the held-out prediction
Direction recovery
SUPPORTING
predicted δv
measured δv
Is the response direction recovered?
Pearson-δ
Allele discrimination
PRIMARY ALLELE-LEVEL SCORE
own δv
predicted δv
siblings
Does δv rank its own δv above same-gene siblings?
PDS
tie-aware percentile rank
chance 0.50 · higher is better
DE-program fidelity
SUPPORTING
measured
predicted
Do changed-gene programs agree?
top-50 DE overlap
DE effect-size Spearman
sign agreement
g
training
held-out
defined only / not scored
bars: realized test n / total n
WITHIN-GENE GENERALIZATION
Random
predefined holdout
T 16/98
K 8/92
G 52/254
J 9/26
Positional
C-terminal half
T 49/98
K 41/92
G 127/254
J 13/26
Mechanistic
hotspot / switch
T 9/98
K 10/92
G 32/254
J 20/26
TRANSFER / DATA-REGIME STRESS TESTS
Cross-gene
GATA1 + JAK1 held out · not scored
T 0/98
K 0/92
G 254/254
J 26/26
Low-depth
<200 cells
T 0/98
K 0/92
G 45/254
J 17/26
Compatibility
prior evaluation protocol
T 10/98
K 8/92
G 52/254
J 0/26